**Pre-Analysis Plan: Many Daughters Project**

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## HYPOTHESIS 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

### Research Hypothesis

**H1:** Having daughters leads to stronger support for gender equality (as measured by attitudes). Formally: (X for respondents with daughters) > (X for respondents without daughters), where X represents a measure of support for gender equality.

### Variables

### Dependent Variable: Gender Equality Attitudes (X)

**Primary measure:** Composite index of gender equality attitudes constructed from SOEP individual questionnaire variables:

- Variables measuring attitudes toward women career aspiration (plh0301)
- Variables measuring attitudes toward men care on financial security (plh0307)
- Additional variables might be included however, they are required to have high correlation above .5 to the first two (after having checked for similar attitudes, none meet the criteria)

**Coding approach:**

- First make sure the cardinality of all variables and scale are equals (ok)
- Individual attitude variables will be standardized (z-scores) before creating composite index
- Composite index will be created using averaging as only two items will be included in the scale
- Higher values indicate stronger support for gender equality
- Missing values handled using listwise deletion for composite index

### Independent Variable: Having Daughters

**Primary measure:** Binary indicator for having at least a daughter

- has_daughter = 1 if respondent has at least a daughter, 0 if any daughters

**Coding approach:**

- No age restrictions on daughters (all daughters counted regardless of current age)

### Control Variables

**Demographic controls:**

- age: Respondent's age (continuous)
- female: Respondent's gender (1=female, 0=male)

- education: Education level in years
- partner: (1= partnered, 0=single)

**Socioeconomic controls:**

- log_hhincome: this will be Log transformed household income
- pgemplst: Employment status (1=employed, 0=not employed) recoded into dummy

**Inclusion/Exclusion Criteria**

**Inclusion criteria:**

- Respondents with non-missing data on gender equality attitudes

**Exclusion criteria:**

- Respondents with inconsistent biographical information about children

**Exact Analysis**

**Regression Model Specification:**

gender_equality_attitudes = $\beta_0 + \beta_1 \times$has_daughter + $\beta_2 \times$age + $\beta_3 \times$female + $\beta_4 \times$education + $\beta_5 \times$ partner+ $\beta_6 \times$ pgemplst+ $\beta_7 \times$log_hhincome + e

**Estimation method:**

- Ordinary Least Squares (OLS) regression with bootstrapped standard errors
- Robust standard errors clustered at household level: vce(cluster cid)

**Interpretation of Results**

**Primary test statistic:** $\beta_1$ (coefficient on has_daughter)

- **Null hypothesis:** $\beta_1 \leq 0$
- **Alternative hypothesis:** $\beta_1 > 0$
- **Test type:** One-tailed t-test
- **Significance level:** $\alpha = 0.05$

**Decision rule:**

- **Support for H1:** $\beta_1 > 0$ AND p-value < 0.05 (one-tailed)
- **No support for H1:** $\beta_1 \leq 0$ OR p-value $\geq 0.05$

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**HYPOTHESIS 2: Having daughters leads to more left-leaning political preferences**

**Research Hypothesis**

**H2:** Having daughters leads to more left-leaning political preferences. Formally: (Y for respondents with daughters) > (Y for respondents without daughters), where Y represents a measure of left-leaning political preferences.

**Variables**

**Dependent Variable: Left-leaning Political Preferences (Y)**

**Primary measure:** Political orientation scale from SOEP individual questionnaire

- Political self-placement on left-right scale (plh0004, 0-10 scale)

**Coding approach:**

- Reverse-code political orientation so higher values indicate more left-leaning preferences
- Recode of missing

**Independent Variable: Having Daughters**

- Same as H1: has_daughter = 1 if respondent has at least a daughter, 0 if no daughters

**Control Variables**

Same control variables as H1, plus:

**Inclusion/Exclusion Criteria**

Same as H1.

**Exact Analysis**

**Regression Model Specification:**

Right_left= $\beta_0 + \beta_1 \times$has_daughter + $\beta_2 \times$age + $\beta_3 \times$female + $\beta_4 \times$education + $\beta_5 \times$ partner+ $\beta_6 \times$ pgemplst+ $\beta_7 \times$log_hhincome + $\varepsilon$

**Estimation method:**

- Ordinary Least Squares (OLS) regression with bootstrapped standard errors
- Robust standard errors clustered at household level: vce(cluster cid)

**Interpretation of Results**

**Primary test statistic:** $\beta_1$ (coefficient on has_daughter )

- **Null hypothesis:** $\beta_1 \leq 0$
- **Alternative hypothesis:** $\beta_1 > 0$
- **Test type:** One-tailed t-test
- **Significance level:** $\alpha = 0.05$

**Decision rule:**

- **Support for H2:** $\beta_1 > 0$ AND p-value < 0.05 (one-tailed)
- **No support for H2:** $\beta_1 \leq 0$ OR p-value $\geq 0.05$

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**HYPOTHESIS 3: Having daughters leads to more gender equality in the household (as measured by behaviours or outcomes) plh0308_v1**

**Research Hypothesis**

**H3:** Having daughters leads to more gender equality in the household (as measured by behaviours or outcomes). Formally: (Z for respondents with daughters) > (Z for respondents without daughters), where Z represents a measure of gender equality in the household.

**Variables**

**Dependent Variable: Equality in the responsibility in household (plh0308_v1)**

**Coding approach:**

- Higher values indicate more gender-equal household arrangements
- Handle missing values using available case analysis
- Reverse coding of plh0308_v1 to indicate higher support for equal distribution of tasks

**Independent Variable: Having Daughters**

- Same as H1 and H2. has_daughter = 1 if respondent has a daughter, 0 if no daughters

**Control Variables**

Same as H1, plus household-specific controls

**Inclusion/Exclusion Criteria**

Same as H1.

**Exact Analysis**

**Regression Model Specification:**

$hh_equality = \beta_0 + \beta_1 \times has_daughter + \beta_2 \times age + \beta_3 \times female + \beta_4 \times education + \beta_5 \times partner + \beta_6 \times pgemplst + \beta_7 \times log_hhincome + \varepsilon$

**Estimation method:**

- Ordinary Least Squares (OLS) regression with bootstrapped standard errors
- Robust standard errors clustered at household level: vce(cluster cid)

**Interpretation of Results**

**Primary test statistic:** $\beta_1$ (coefficient on has_daughter )

- **Null hypothesis:** $\beta_1 \leq 0$
- **Alternative hypothesis:** $\beta_1 > 0$
- **Test type:** One-tailed t-test
- **Significance level:** $\alpha = 0.05$

**Decision rule:**

- **Support for H3:** $\beta_1 > 0$ AND p-value < 0.05 (one-tailed)
- **No support for H3:** $\beta_1 \leq 0$ OR p-value $\geq 0.05$

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**HYPOTHESIS 4: Having daughters leads to a larger increase in support of gender equality for men than for women**

**Research Hypothesis**

**H4:** Having daughters leads to a larger increase in the support of gender equality for men than for women. Formally: (X for male respondents with daughters - X for male respondents without daughters) > (X for female respondents with daughters - X for female respondents without daughters)

**Variables**

**Dependent Variable: Gender Equality Attitudes (X)**

Same as H1

**Independent Variables**

- has_daughter = 1 if respondent has at least a daughter, 0 if no daughters
- female: Respondent's gender (1=female, 0=male)
- has_daughter _x_female: Interaction term between has_daughter and female

**Control Variables**

Same as H1

**Inclusion/Exclusion Criteria**

Same as H1

**Exact Analysis**

**Regression Model Specification:**

$gender_equality_attitudes = \beta_0 + \beta_1 \times has_daughter + \beta_2 \times female + \beta_3 \times has_daughter_x_female + \beta_4 \times age + \beta_5 \times education + \beta_6 \times partner + \beta^7 \times pgemplst + \beta_8 \times log_hhincome + \varepsilon$

**Key coefficients interpretation:**

- $\beta_1$: Effect of having a daughter for men (reference group)
- $\beta_1 + \beta_3$: Effect of having a daughter for women
- $\beta_3$: Difference in daughter effect between women and men

**Interpretation of Results**

**Primary test statistic:** $-\beta_3$ (negative of coefficient on interaction term)

**Hypothesis test:**

- **Null hypothesis:** $\beta_3 \geq 0$ (effect is equal or larger for women)
- **Alternative hypothesis:** $\beta_3 < 0$ (effect is larger for men)
- **Test type:** One-tailed t-test on $-\beta_3$
- **Significance level:** $\alpha = 0.05$

**Decision rule:**

- **Support for H4:** $\beta_3 < 0$ AND p-value < 0.05 (one-tailed test of $-\beta_3 > 0$)
- **No support for H4:** $\beta_3 \geq 0$ OR p-value $\geq 0.05$

**Additional interpretation:**

- Calculate marginal effects:
    - Men's daughter effect = $\beta_1$
    - Women's daughter effect = $\beta_1 + \beta_3$
    - Difference = $\beta_1 - (\beta_1 + \beta_3) = -\beta_3$

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**General Analysis Specifications**

**Data Management**

- **SOEP datasets**
- **Analytical approach:** Cross-sectional analysis

**Statistical Software**

- **Primary software:** Stata 15
- **Standard errors:** Robust standard errors clustered at household level for all models
- **Missing data:** Listwise deletion for main analyses

**Robustness Checks**

1. Alternative measures of key variables when available
2. Different sample restrictions (e.g., only parents with 2+ children)
3. Sensitivity to control variable specifications

**Limitations and Considerations**

**Causal Inference**

This analysis relies on observational data. While we treat these as causal hypotheses, identification assumes having daughters is quasi-random conditional on total number of children and other controls.

**Potential Confounders**

- Fertility preferences and family planning decisions
- Unobserved parental characteristics
- Time-varying factors affecting both fertility and attitudes

**Data Quality**

- Reliance on self-reported measures for attitudes and behaviours
- Potential measurement error in key variables
- Missing data patterns may introduce selection bias

**External Validity**

Results apply to the German context for a specific year.